

A Passive Threshold-Voltage-Mismatch Cancellation Technique for Visual-Neuron-Inspired HDR CMOS Image Sensors

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Abstract—This paper addresses the critical issue of high fixed-pattern noise (FPN) caused by source-follower threshold-voltage (V_{th}) mismatch in high-dynamic-range (HDR) CMOS image sensor (CIS) inspired by visual neuronal response. We propose a threshold-voltage-mismatch cancellation technique that employs a simple capacitor-based sampling network to dynamically store and cancel pixel-specific V_{th} offsets during the quantization phase. This approach effectively suppresses FPN without requiring power-intensive operational amplifiers or complex digital circuitry. Pre-layout simulation results in a 180 nm CMOS process demonstrate that under significant process induced V_{th} variations, the proposed technique reduces the standard deviation of output digital codes from 11.56 LSB to 1.04 LSB, achieving 91.0% suppression of V_{th} -induced FPN. The circuit architecture introduces zero static power overhead, making it ideally suited for high-quality bio-inspired vision sensing in ultra-low-power Internet of Things applications.

Keywords—CMOS Image Sensor(CIS), Fixed-Pattern Noise (FPN), Threshold Voltage Mismatch, Low-Power Circuit

I. INTRODUCTION

The pursuit of high-fidelity visual information under varying illumination conditions continues to drive the advancement of high-dynamic-range (HDR) imaging technology. CMOS image sensors (CIS) capable of capturing scenes with extreme illumination variations have become increasingly critical for application domains such as autonomous driving, security surveillance, and power-constrained Internet of Things devices[1][2]. Conventional multi-exposure HDR techniques extend dynamic range by synthesizing multiple frames captured with different exposure times. However, this "synthesize-then-compress" paradigm requires complex off-chip processing, introduces motion artifacts, and results in high power consumption, making it unsuitable for always-on, resource-constrained applications[3][4]. Single-frame HDR techniques based on nonlinear compression have emerged as a promising alternative to circumvent these issues. These approaches, including logarithmic response[5] and piecewise linear response[6], directly compress the wide-ranging scene illumination within the pixel or during readout. Nevertheless, they typically suffer from significant fixed-pattern noise (FPN), limited output swing, and poor signal-to-noise ratio under low-light conditions[7].

Inspired by the human visual system, bio-inspired designs that mimic the nonlinear response of visual neurons offer a viable pathway toward low-power HDR imaging. Previous work[8] introduced a novel quantization scheme

based on the Naka-Rushton function, successfully achieving single-exposure HDR with low power consumption. Although this sensor achieved a dynamic range exceeding 88.96 dB, it inherently suffers from high FPN due to the absence of power-intensive correlated double sampling (CDS) techniques. The primary source of this FPN is the threshold-voltage mismatch in the in-pixel source-follower amplifier, which introduces spatially fixed offsets across the sensor array. Traditional FPN suppression techniques, particularly CDS, while effective, come at a high cost. Analog-domain CDS circuits based on operational amplifiers typically dominate a significant portion of the analog power budget[9]. Although digital-domain CDS offers greater flexibility, it requires complex timing control and additional digital logic, indirectly increasing both power consumption and design complexity[10]. Consequently, there is a pressing need for a low-power FPN reduction technique that is inherently compatible with nonlinear quantization schemes.

To address this critical challenge, this paper presents a novel passive FPN reduction technique specifically designed for Naka-Rushton function-based HDR CIS. The proposed technique uses a simple capacitor-based sampling network to dynamically store and subtract pixel-specific V_{th} offsets during the quantization phase. Its passive nature ensures zero static power overhead, making it an ideal choice for low-power applications. The core innovation of this technique lies in its ability to make the comparator's switching moment independent of the source-follower's V_{th} mismatch, without requiring active amplifiers or complex digital calibration. Post-layout simulation results in a standard 180 nm CMOS process verify the functionality of the proposed circuit. Analysis of the test results demonstrates that the technique can reduce V_{th} -induced FPN by over 90%.

The remainder of this paper is organized as follows. Section II provides a brief overview of the visual-neuron-inspired HDR image sensor operating principle and the origin of its inherent FPN, followed by a detailed explanation of the architecture and working principle of the proposed passive FPN reduction technique. Section III presents the simulation results and expected performance. Finally, Section IV concludes the paper.

II. CIRCUIT ARCHITECTURE AND OPERATING PRINCIPLE

The visual-neuron-inspired HDR image sensor enables simultaneous pixel exposure and analog-to-digital conversion through modified control signals and a revised conversion methodology. This allows integrating-type pixels to achieve a nonlinear response, where the proposed nonlinear response characteristic conforms to a linear transformation of the Naka-Rushton equation with a Hill

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coefficient of $n = 1$. Consequently, this image sensor possesses bio-inspired properties that emulate the nonlinear response of biological visual systems.

Fig. 1 illustrates the cascaded architecture implementing the nonlinear response, based on an integrating-type 3T pixel and a single-slope (SS) ADC.

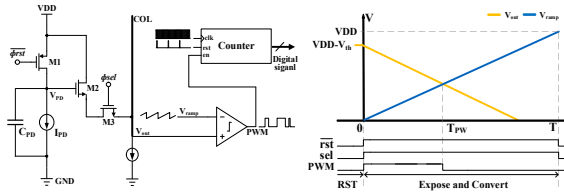


Fig. 1. The cascaded structure of the integrating-type 3T pixel and the SS ADC.

In this architecture, during the reset phase, transistor M1 pulls the V_{PD} node up to V_{DD} and turns off at time zero, leaving the photodiode (PD) node floating. Simultaneously, M3 turns on, connecting the output voltage (V_{out}) of the source follower M2 to the column bus. Throughout the exposure period from 0 to T , V_{PD} gradually decreases from V_{DD} due to the accumulation of photogenerated charge, while V_{out} , being offset from V_{PD} by the V_{th} of M2, starts from $V_{DD} - V_{th}$ and decreases accordingly. Concurrently, the ramp voltage (V_{ramp}) increases linearly from 0 to the supply voltage V_{DD} over the same period. V_{out} is directly connected to the non-inverting input of the comparator without a sample-and-hold circuit, resulting in both comparator inputs being dynamically varying voltage signals. Since V_{out} starts at $V_{DD} - V_{th}$ and decreases while V_{ramp} ramps up from 0 to V_{DD} , there must exist one and only one time instant within the period where the pixel output V_{out} equals the V_{ramp} . This equality causes the

comparator output to flip polarity, thereby generating a pulse-width modulation (PWM) signal.

The expression for V_{out} as a function of time is given by:

$$V_{out} = V_{DD} - V_{th} - \frac{I_{PD} \cdot t}{C_{PD}} \quad (1)$$

The V_{ramp} as a function of time is expressed as:

$$V_{ramp} = V_{DD} * \frac{t}{T} \quad (2)$$

Where V_{DD} is the supply voltage, V_{th} is the threshold voltage of M2, I_{PD} is the photocurrent, C_{PD} is the photodiode junction capacitance, and T is one exposure period.

Consequently, the pulse-width signal, which encapsulates the illumination information, can be derived as:

$$T_{PW} = \frac{(V_{DD} - V_{th}) * T * C_{PD}}{I_{PD} * T + V_{DD} * C_{PD}} \quad (3)$$

For different pixels:

$$T_{PW1} = \frac{V_{DD} - V_{th1}}{\frac{V_{DD} - V_{th1}}{T} + \frac{I_{PD1}}{C_{PD}}} \quad (4)$$

$$T_{PW2} = \frac{V_{DD} - V_{th2}}{\frac{V_{DD} - V_{th2}}{T} + \frac{I_{PD2}}{C_{PD}}} \quad (5)$$

Therefore, the T_{PW} contains the photocurrent information I_{PD} and is influenced by the V_{th} . For the same illumination intensity I_{PD} , differences in V_{th} will cause variations in T_{PW} , thereby introducing errors during the quantization phase. Since manufacturing process variations lead to mismatch in the V_{th} of transistor M2 across different pixels, this V_{th} variation causes FPN in T_{PW} , ultimately resulting in

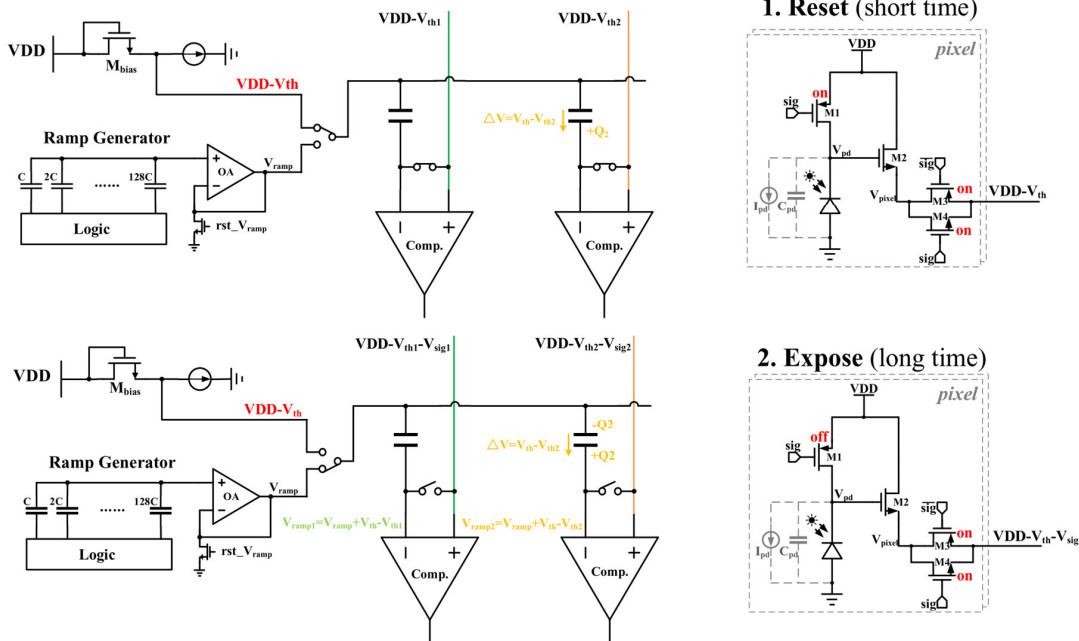


Fig. 2. Circuit diagram of the passive noise reduction technique.

quantization errors.

To address this issue, we propose a novel passive V_{th} variation cancellation technique for the implemented quantization method. Its operation does not rely on operational amplifiers, thereby achieving noise suppression with low power consumption. The implementation steps and operating principle of this technique are illustrated in Fig. 2.

In this method, V_{th1} and V_{th2} represent the threshold voltages of the source-follower M2 in Pixel 1 and Pixel 2, respectively.

The operating sequence is divided into two phases:

- During the pixel reset phase, $VDD - V_{th2}$ is sampled onto the bottom plate of the capacitor, while the top plate is connected to a globally shared bias. This bias, regulated by transistor M_{bias} (composed of multiple parallel-connected MOSFETs identical to the in-pixel M2), provides a $VDD - V_{th}$ like voltage drop, minimizing leakage current by maintaining a consistent voltage difference $V_{th} - V_{th2}$ across the capacitor plates.
- In the exposure and comparison phase, the sampled voltage difference is superimposed onto the V_{ramp} . As shown in the waveform diagram below, for pixels with different V_{th} values, the input waveforms at the two terminals of each comparator are shifted solely along the y-axis.

$$V_{ramp1} = V_{ramp} + V_{th} - V_{th1} \quad (6)$$

$$V_{ramp2} = V_{ramp} + V_{th} - V_{th2} \quad (7)$$

Therefore, T_{PW1} and T_{PW2} are expressed as:

$$T_{PW1} = \frac{VDD - V_{th}}{\frac{VDD}{T} + \frac{I_{PD1}}{C_{PD}}} \quad (8)$$

$$T_{PW2} = \frac{VDD - V_{th}}{\frac{VDD}{T} + \frac{I_{PD2}}{C_{PD}}} \quad (9)$$

The comparator's switching point becomes independent of the pixel-specific V_{th} mismatch, thereby effectively eliminating FPN.

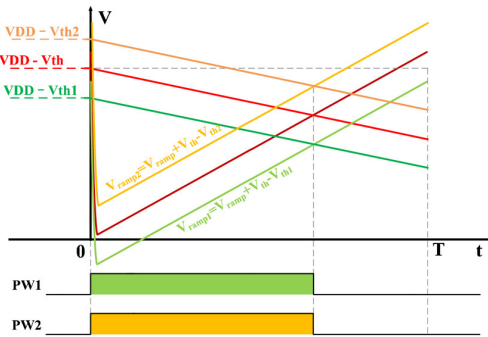


Fig. 3. Operating waveforms of the proposed passive FPN reduction technique.

III. SIMULATION AND VERIFICATION

To validate the effectiveness of the proposed passive FPN reduction technique, we conducted comprehensive circuit simulations. The primary objective of these simulations was to evaluate the technique's capability in improving the consistency of the system's final digital output code under different threshold voltage (V_{th}) conditions.

Since the threshold voltage (V_{th}) of a transistor is a function of its body bias voltage (V_B), following the body effect formula:

$$V_{th} = V_{th0} + \gamma(\sqrt{|2\phi_F + V_{SB}|} - \sqrt{|2\phi_F|}) \quad (10)$$

where V_{th0} is the threshold voltage at zero body bias, γ is the body effect coefficient, and ϕ_F is the Fermi potential.

To emulate process-induced V_{th} mismatch, we swept the V_B of the in-pixel source follower (M2). In our simulations, V_B was scanned from 0V to -1V to simulate a wide range of V_{th} variations. Based on estimation in a 180nm CMOS process, the variation of V_{th} corresponding to V_B is approximately 200mV, which adequately covers the V_{th} variation range induced by process variations. Under fixed uniform illumination conditions with a constant photocurrent I_{pd} , we recorded both the comparator's output pulse width and the final 8-bit digital code from the latch circuit, before and after applying the proposed cancellation technique.

In Fig. 4, the comparator output waveforms corresponding to different V_B values (different V_{th}) exhibit significant dispersion along the time axis. The falling-edge timing of each curve, which represents the comparator's switching point, shows clear shifts as V_B varies. This phenomenon confirms that the threshold voltage mismatch of the pixel source follower directly introduces variations in the quantization timing, consequently generating fixed brightness differences in the final image.

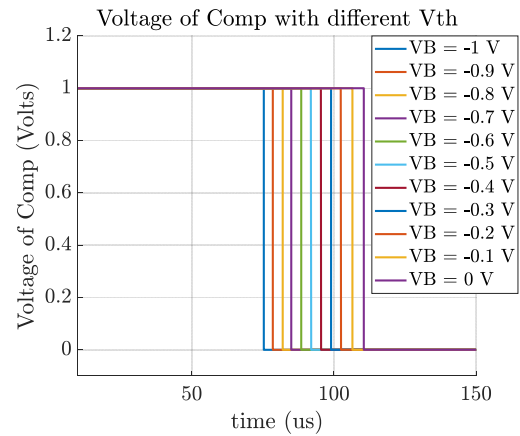


Fig. 4. Comp Voltage without FPN reduction.

In stark contrast to Fig. 4, after applying the proposed cancellation technique, Fig. 5 shows that all curves align almost perfectly despite the same wide variation in V_B . The comparator's switching timing remains consistent and becomes independent of V_{th} variations. This visually demonstrates, at the analog level, the successful isolation of

V_{th} mismatch from the signal path by the proposed technique, thereby ensuring uniformity in the quantization process.

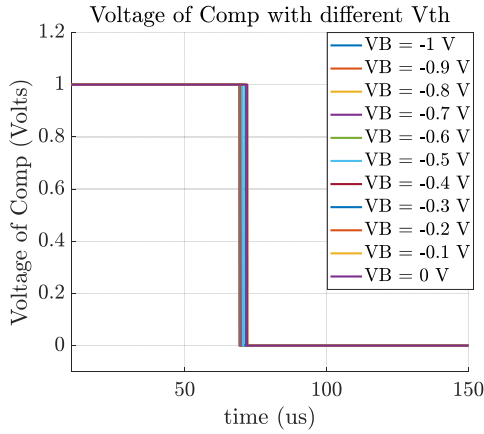


Fig. 5. Comp Voltage with FPN reduction.

Fig. 6 quantitatively reveals the severity of FPN in the original design. As V_B varies, the output digital code fluctuates drastically within a wide range from 78 to 110, spanning up to 32 LSB. Statistical results show a standard deviation σ of 11.56 LSB for the output codes. Significant variation indicates that under uniform illumination, pixels with different V_{th} would exhibit substantially different output brightness, resulting in severe FPN in the final image.

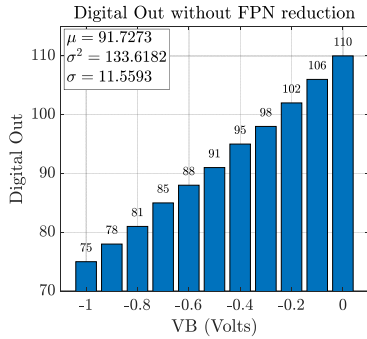


Fig. 6. Digital Out without FPN reduction.

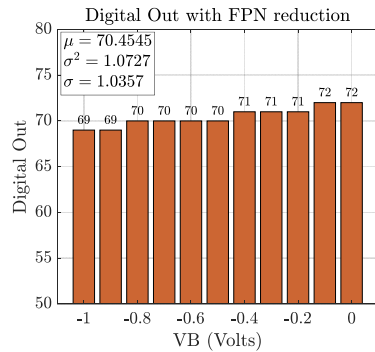


Fig. 7. Digital Out with FPN reduction.

Fig. 7 shows the improvement in digital output after applying the proposed cancellation technique is remarkable. The output codes are stabilized around 70, with the

fluctuation range drastically compressed. The standard deviation σ is dramatically reduced from the original 11.56 LSB to merely 1.04 LSB.

To quantify the improvement, we calculated the reduction ratio of the output code standard deviation:

$$FPN_{\text{Suppression Rate}} = \left(1 - \frac{\sigma_{\text{new}}}{\sigma_{\text{prior}}}\right) \times 100$$

$$= \left(1 - \frac{1.04}{11.56}\right) \times 100\% \approx 91.0\% \quad (11)$$

The results demonstrate that the proposed technique successfully suppresses 91% of the output code fluctuation induced by V_{th} mismatch. During the per-pixel readout phase, the proposed technique achieves a 50% reduction in power consumption compared to the conventional CDS circuit.

TABLE I. COMPARISON TABLE WITH OTHER WORKS

	<i>This work</i>	<i>SSC-L '24 [11]</i>	<i>TCAS-I '20 [12]</i>	<i>JSSC '25 [4]</i>
Process	180nm CMOS	65/40 Stacked	130nm CMOS	350nm CIS
Pixel Pitch	6.6 μm	6.3 μm	6.5 μm	7.2 μm
Frame Rate	60 fps	30 fps	100 fps	60 fps
ADC	8-bit	10-bit	10-bit	10-bit
DR (dB)	>88.96	120	N/A	90.5
FPN (%)	0.99(prior) 0.089(new)	0.6	1.41	0.32
HDR Tech.	Visual Neuronal Response (VNR)	Logarithmic counter & ramp	Pixel-wise exposure	Adaptive DCG

Table I compares the performance of the proposed CIS with state-of-the-art single-frame HDR techniques. The proposed FPN reduction technique not only enables operation with high dynamic range but also maintains low FPN, achieving high-quality, high-dynamic-range, and low-power imaging.

IV. CONCLUSION

This paper has presented a novel passive threshold-voltage-mismatch cancellation circuit that effectively suppresses FPN induced by source-follower V_{th} mismatch in HDR image sensors, serving as a viable replacement for power-hungry operational-amplifier-based CDS. The proposed method requires no operational amplifiers and introduces negligible power and area overhead. Post-layout simulation results have confirmed its functionality. This technique significantly enhances the suitability of ultra-low-power VNR-type CIS for high-quality imaging in resource-constrained.

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